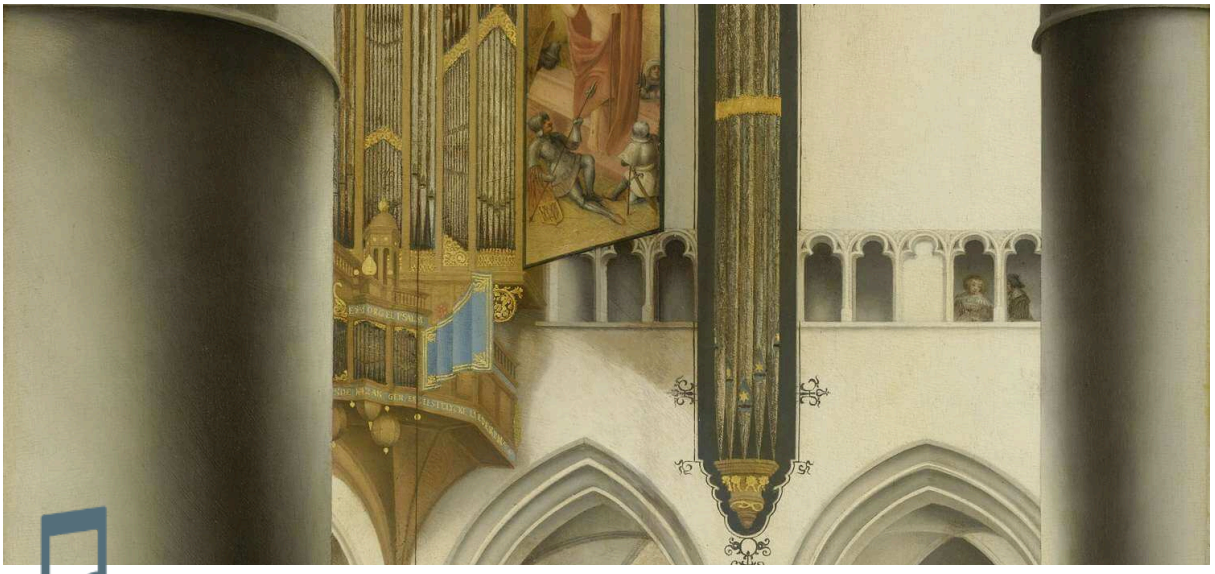


2022/2021 Maths paper

1. Is the following proposition true or false? Justify your answer fully.

   If 2n+7 is prime then n is prime.

   let's the the contrapositive

   "if n is not prime, then 2n+7 is not prime"

   n = 6 (not prime) →2(6) + 7 = 19 (prime)

   the proposition is false when n = 6

2. Prove that the following statement is true for all integers n:

   "n + 3 is odd if and only if 3n is a multiple of 6."
   n + 3 is odd when 3n is a multiple of 6, also 3n is a multiple of 6 when n + 3 is odd.

   if n+3 is odd, we can rewrite it as 2k+1, for all k (integer)

$$n + 3 = 2k + 1$$
$$n = 2k - 2$$

   assuming n+3 is odd, 3n can written as 6l.

$$3(2k-2) =$$
$$6k-6 =$$
$$6(k-1) =$$
$$6l, "for\ some\ l \in \mathbb{Z}$$

6l is a multiple of 6, when n+3 is odd, 3n will be a multiple of 6

next case:

let 3n = 6m, for m (integer)

n = 2m.

$$n+3 =$$
$$2m+3 =$$
$$2m+2+1 =$$
$$2(m+1)+1 =$$
$$2p+1$$
$$for\ p \in \mathbb{Z}$$

since n+3 can be written as 2p+1, it has to be odd.

3. hi
4. Solve each of the following pairs of modular equations (simultaneous congruences), if possible. If no solution exists, explain why not.

   (i) Find integers, x and y, such that
   3x+5y ≡ 14(mod28)

   5x+9y ≡ 6(mod28)
   (ii) Find integers, x and y, such that

   x+5y ≡ 3(mod9)
   4x+5y ≡ 1(mod9)

   (iii) Find a positive integer, x, such that both
   x ≡ 6(mod8)

   x ≡ 17 (mod 25)
   State clearly what method you use to find your solution.

   answer:

   (i)